

Practical – 4

Practical Title: JavaScript DOM Manipulation, Event Handling, and UI Interactivity

Problem Definition: Add dynamic UI components such as collapsible FAQ, modal popup, image/content slider, notification banner, hamburger menu, and light/dark theme switcher using JavaScript.

A. Theory

1. How are DOM elements selected and modified?
The StudentHub Practical 4 script selects elements using methods such as `getElementById()` and `querySelectorAll()`. After selecting an element, JavaScript changes classes, attributes, text, visibility, and styles. For example, the FAQ uses class changes to show or hide answers, while the theme switcher adds or removes the dark-mode class from the body.
2. Are event listeners attached correctly?
Event listeners are attached after the DOM is ready using `DOMContentLoaded`. Click and input events are used for the hamburger menu, theme button, FAQ questions, modal controls, slider buttons, notification controls, assignment search, and form validation. The script also checks whether optional elements exist before attaching listeners, so pages without a particular component do not produce unnecessary JavaScript errors.
3. Is `localStorage` used for remembering theme choice?
Yes. The selected theme is stored with the key `studentHubTheme`. When a page loads, JavaScript reads the saved value and applies the dark-mode class when the saved preference is dark. When the theme button is used, the new preference is saved again. Therefore, the theme preference can be restored on later page loads in the same browser context.
4. Does interactivity improve usability without breaking accessibility?
The interactive features are designed to keep the interface usable. Buttons have meaningful labels, the hamburger menu updates `aria-expanded`, the active navigation page uses `aria-current`, FAQ buttons expose their expanded state, and the modal uses dialog-related ARIA attributes. The modal also restores focus to the element that opened it and can be closed with the Escape key.

B. Implementation/ Output/Result:

The existing StudentHub portal was enhanced with a common JavaScript file js/script.js. The same interactive structure is used across the portal pages while page-specific code runs only when the required element is present.

Feature	Implementation in StudentHub
DOM Manipulation	getElementById(), querySelectorAll(), classList, attributes, textContent and hidden state.
Event Handling	click, input, submit, DOMContentLoaded and keydown events.
Hamburger Menu	Toggles the navigation menu and updates aria-expanded/aria-label.
Theme Switcher	Toggles dark-mode and stores the choice in localStorage.
FAQ Accordion	Opens one answer at a time and updates aria-expanded.
Modal Popup	Opens/closes the dialog, manages focus, supports Escape and overlay closing.
Content Slider	Previous/next controls plus automatic slide rotation.
Notification Banner	Dismissible banner and notification counter reset.
Assignment Search	Filters assignment cards as the user types.
Form Validation	Uses HTML constraint validation and additional password checking.

Important JavaScript Implementation :

- Theme persistence: the script reads and writes the studentHubTheme key using localStorage and restores the dark-mode class during page load.
- FAQ: each .faq-question receives a click listener. The selected answer receives the show class while other answers are closed.
- Slider: slides are collected with querySelectorAll('.slide'). The active slide is changed by class manipulation, with previous/next controls and a five-second automatic rotation.
- Modal: the dialog is opened by changing its hidden state, focus is moved to the close button, and Escape/overlay/close controls can dismiss it.
- Page safety: optional page-specific components are checked before listeners are attached. This allows the common script to run across all StudentHub pages.

C. Evidence of Work (Screenshots):

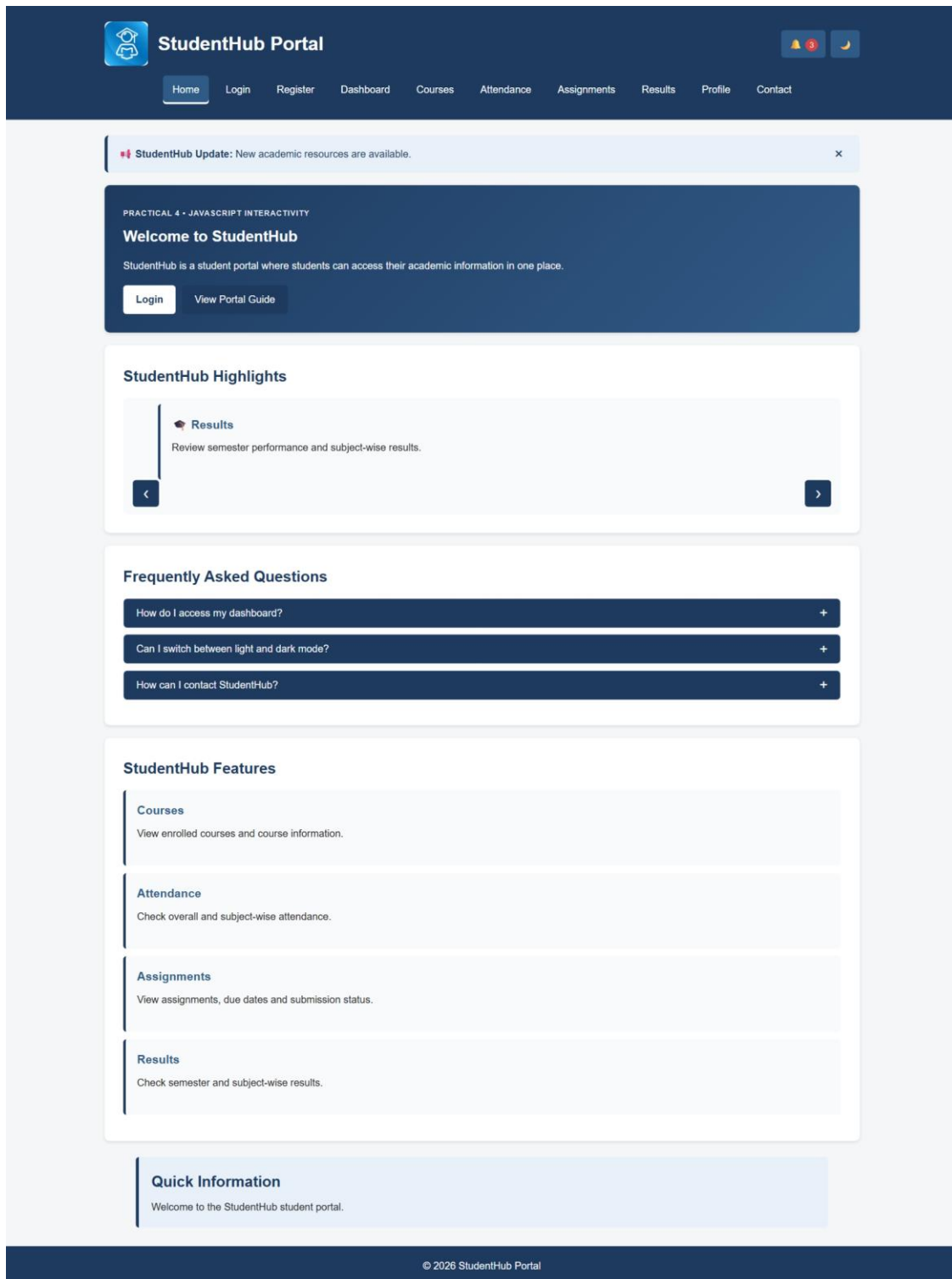
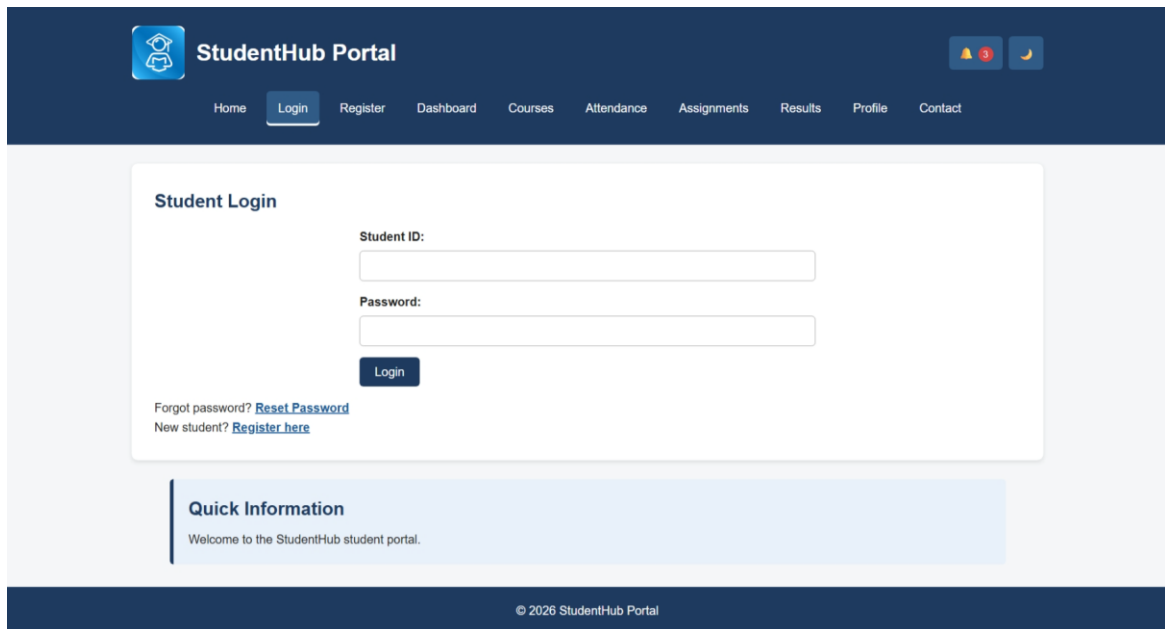


Fig1: Index



The image shows the login page of the StudentHub Portal. The header is dark blue with the StudentHub logo and name on the left, and notification and theme icons on the right. A navigation bar below the header contains links for Home, Login, Register, Dashboard, Courses, Attendance, Assignments, Results, Profile, and Contact. The main content area features a 'Student Login' form with fields for Student ID and Password, a Login button, and links for 'Forgot password? Reset Password' and 'New student? Register here'. Below the form is a 'Quick Information' section with a welcome message. The footer is dark blue with the copyright notice '© 2026 StudentHub Portal'.

StudentHub Portal

Home Login Register Dashboard Courses Attendance Assignments Results Profile Contact

Student Login

Student ID:

Password:

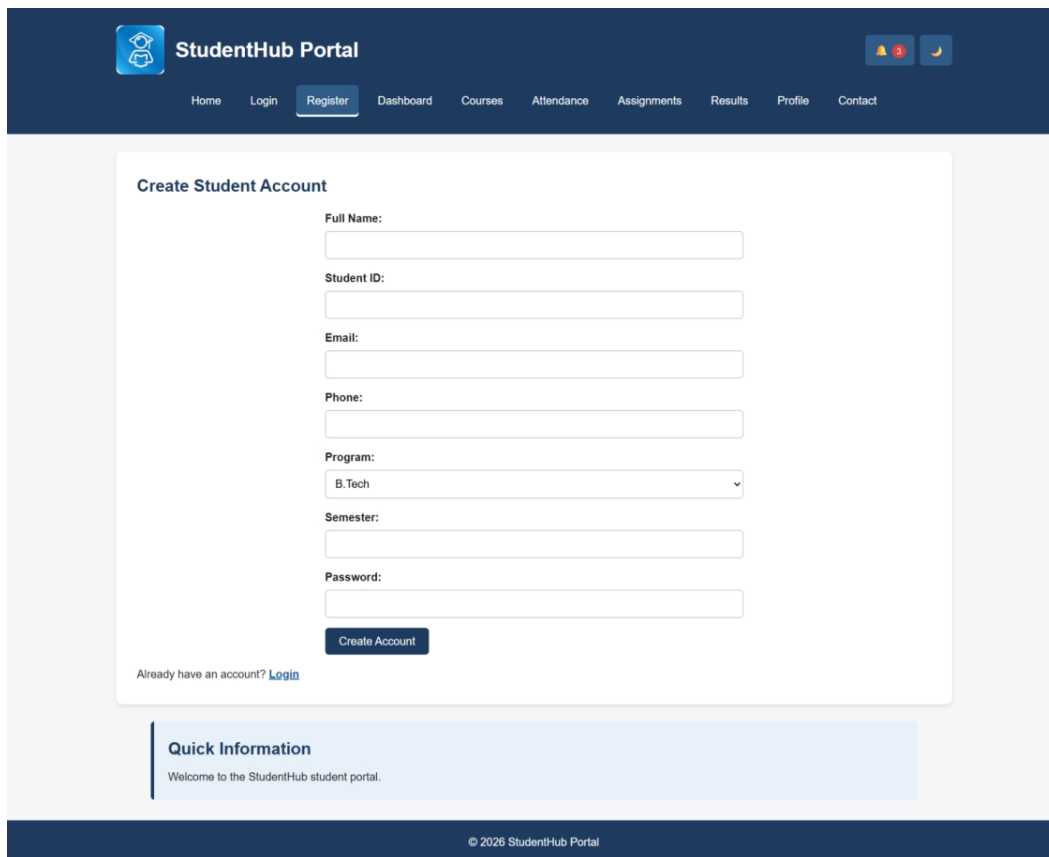
Login

Forgot password? [Reset Password](#)
New student? [Register here](#)

Quick Information
Welcome to the StudentHub student portal.

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Fig2: Login



The image shows the register page of the StudentHub Portal. The header and navigation bar are identical to the login page. The main content area features a 'Create Student Account' form with fields for Full Name, Student ID, Email, Phone, Program (a dropdown menu currently showing 'B.Tech'), Semester, and Password. There is a 'Create Account' button and a link for 'Already have an account? Login'. Below the form is a 'Quick Information' section with a welcome message. The footer is dark blue with the copyright notice '© 2026 StudentHub Portal'.

StudentHub Portal

Home Login Register Dashboard Courses Attendance Assignments Results Profile Contact

Create Student Account

Full Name:

Student ID:

Email:

Phone:

Program:

Semester:

Password:

Create Account

Already have an account? [Login](#)

Quick Information
Welcome to the StudentHub student portal.

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Fig3: Register

StudentHub Portal

Home Login **Register** Dashboard Courses Attendance Assignments Results Profile Contact

Create Student Account

Full Name:

Student ID:

Email:

Phone:

Program:

Semester:

Password:

[Create Account](#)

Already have an account? [Login](#)

Quick Information

Welcome to the StudentHub student portal.

© 2026 StudentHub Portal

Fig4: Dashboard

StudentHub Portal

Home Login Register Dashboard **Courses** Attendance Assignments Results Profile Contact

My Courses

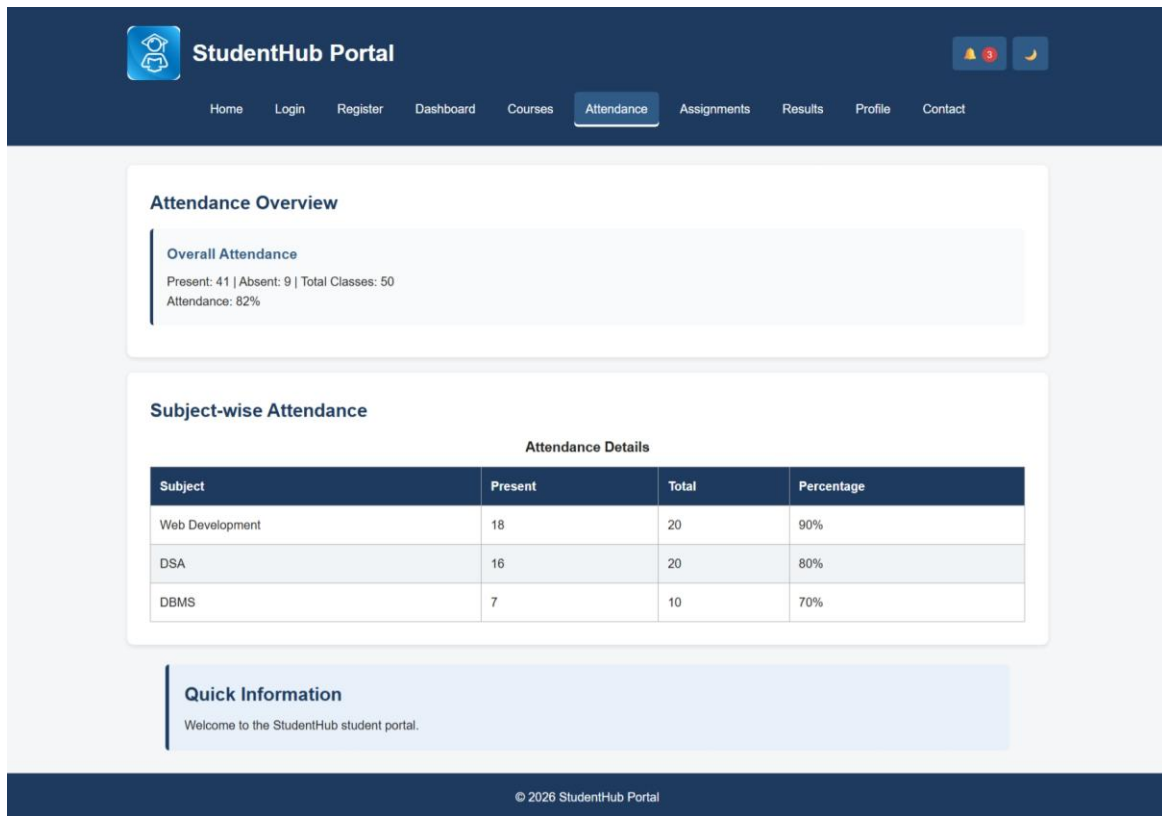
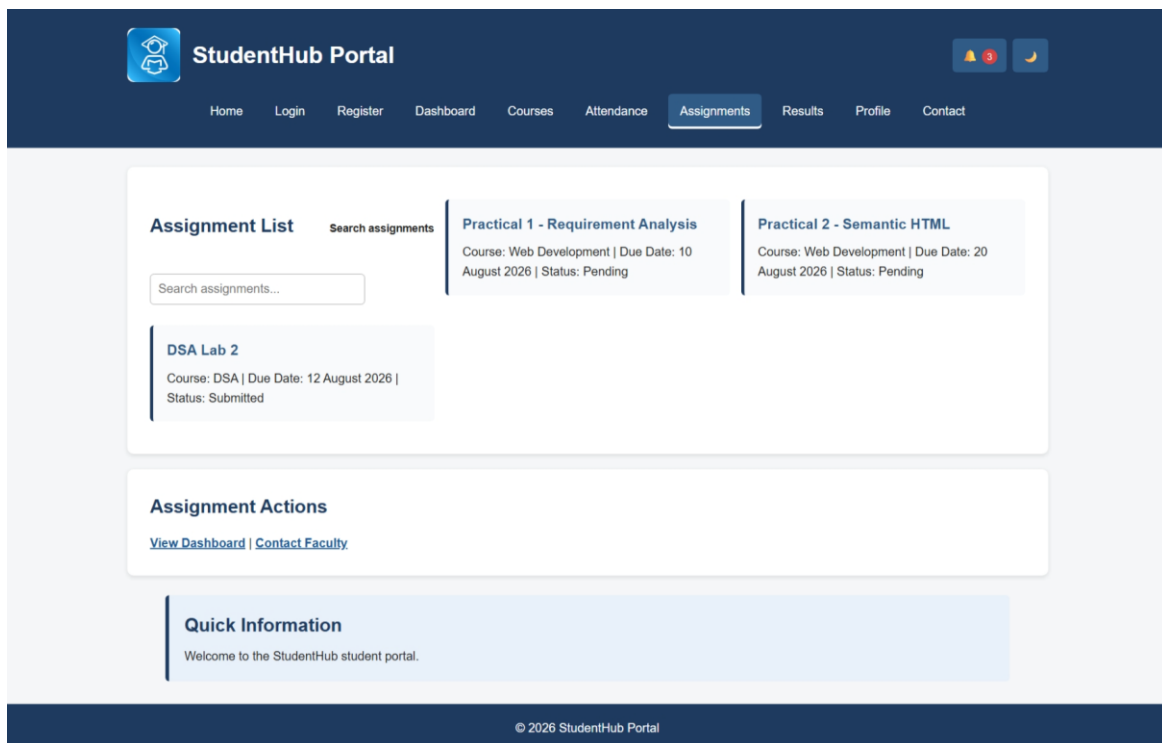
Web Development Course Code: CS301 Faculty: Prof. Sharma Credits: 4	Data Structures and Algorithms Course Code: CS302 Faculty: Prof. Patel Credits: 4	Database Management System Course Code: CS303 Faculty: Prof. Shah Credits: 4
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Quick Information

Welcome to the StudentHub student portal.

© 2026 StudentHub Portal

Fig5: Courses

**Fig6: Attendance****Fig7: Assignment**

The screenshot shows the 'Results' page of the StudentHub Portal. The header includes the portal logo, navigation links (Home, Login, Register, Dashboard, Courses, Attendance, Assignments, Results, Profile, Contact), and notification icons. The main content area features a 'Semester Result' section with current semester statistics (Semester: 3, SGPA: 8.6, CGPA: 8.4) and a 'Subject Results' section displaying a table of Semester 3 results for Web Development, DSA, and DBMS subjects. A 'Quick Information' section at the bottom provides a welcome message.

Semester 3 Results			
Subject	Internal	External	Grade
Web Development	42	46	A
DSA	40	44	A
DBMS	41	43	A

Fig8: Result

The screenshot shows the 'Profile' page of the StudentHub Portal. The header is identical to the previous page, but the 'Profile' link is highlighted in the navigation bar. The main content area features a 'Student Profile' section with personal and academic details, 'Profile Actions' with links to update the profile and login, and a 'Quick Information' section at the bottom.

Student Profile	
Personal Details Name: Aryan Chauhan Student ID: STU001 Email: student@example.com Phone: 9876543210	Academic Details Program: B.Tech Department: Computer Engineering Semester: 3

Fig9: Profile

**StudentHub Portal**



HomeLoginRegisterDashboardCoursesAttendanceAssignmentsResultsProfileContact

Contact StudentHub

If you have questions or problems with the portal, contact us using the form below.

Name:

Email:

Subject:

Message:

Send Message

Quick Information

Welcome to the StudentHub student portal.

© 2026 StudentHub Portal

Fig10: Contact

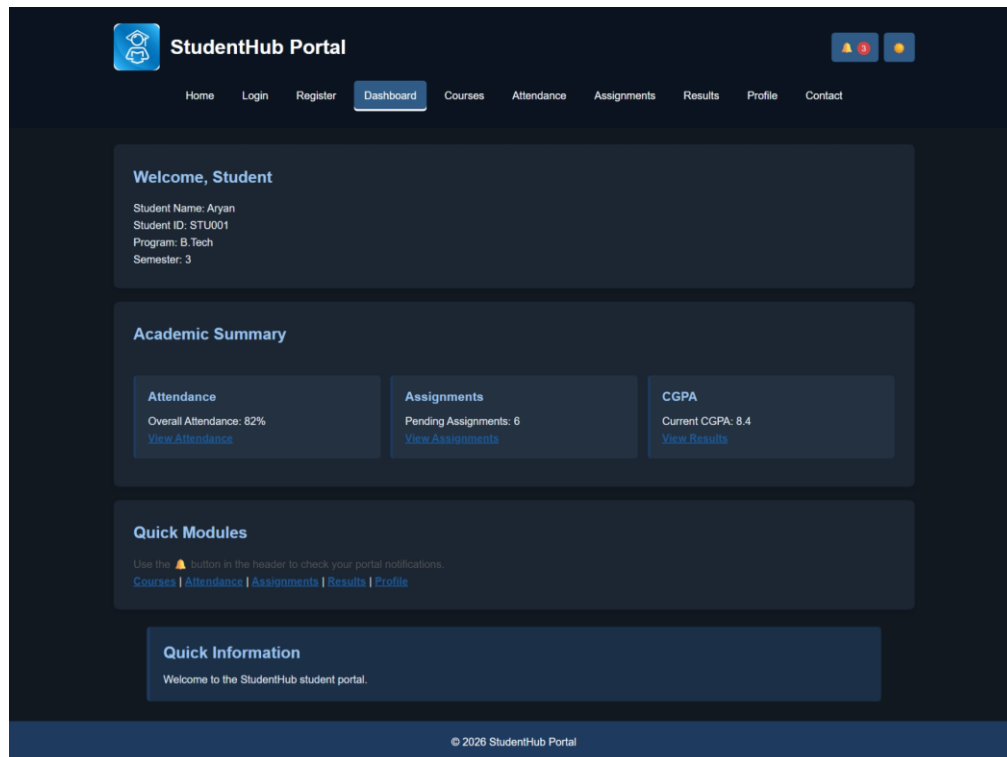


Fig11: Light/Dark Theme Switcher

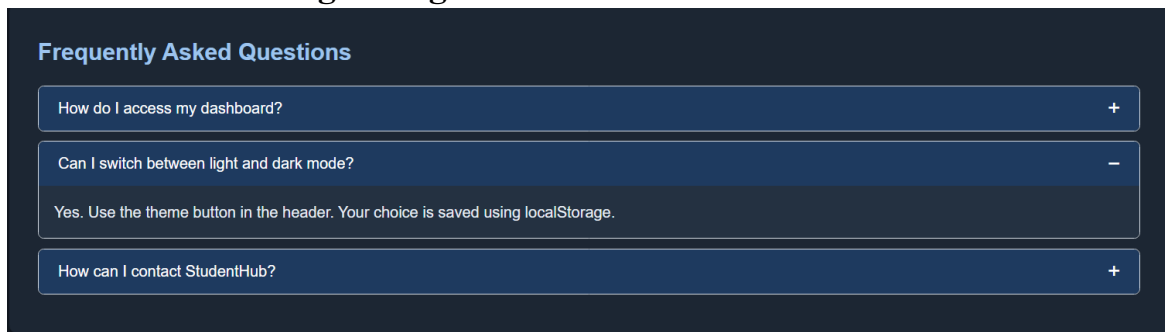


Fig12: FAQ Accordion

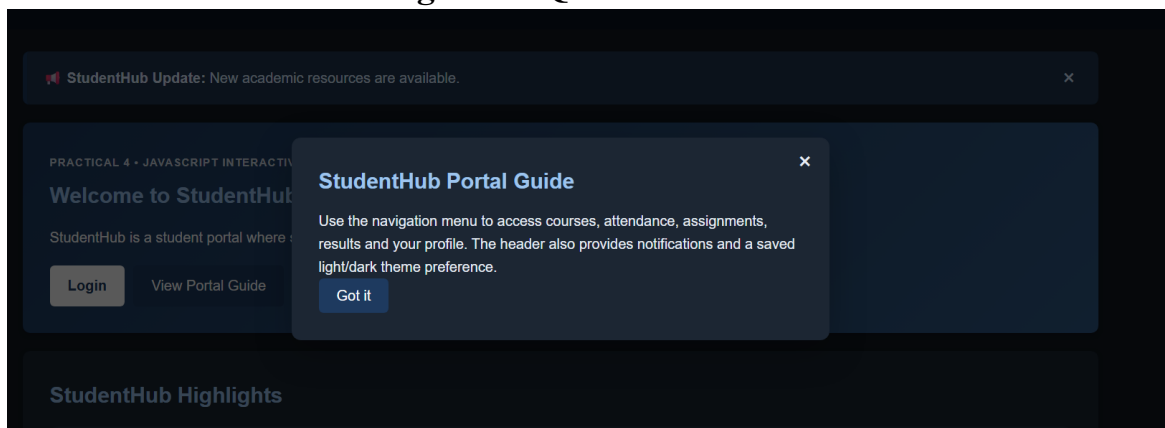


Fig13: Modal Popup

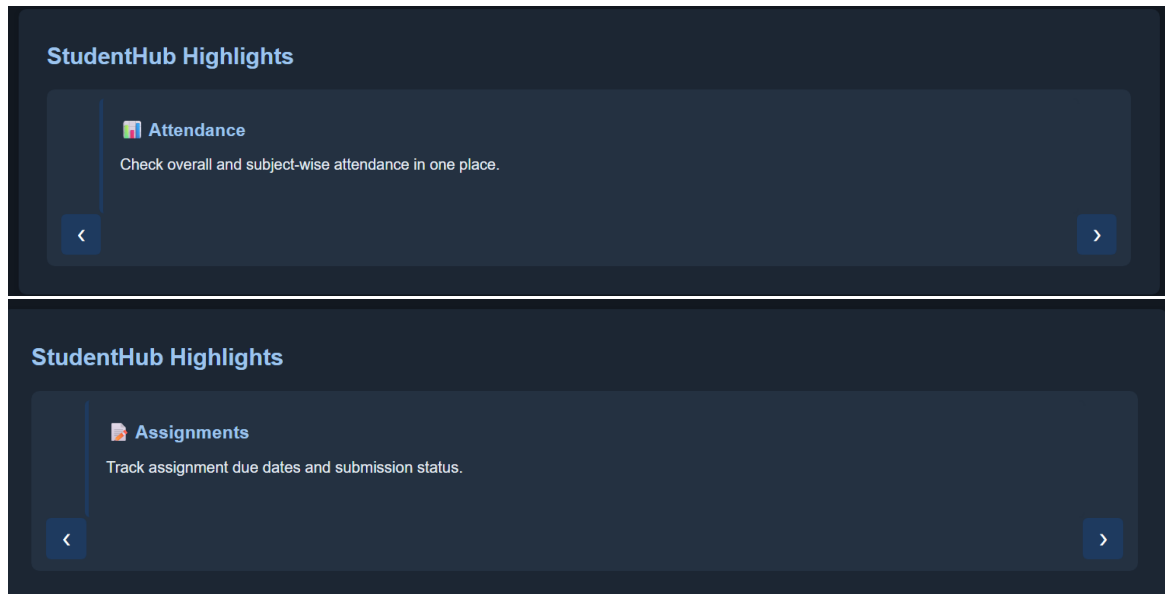


Fig14: Content Slider

D. GitHub Repository Link:

[aryan150806-wq/StudentHubPortal: Web Development Framework P](https://github.com/aryan150806-wq/StudentHubPortal)

E. Challenges Faced:

- Understanding DOM selection and applying changes to different elements without affecting other pages.
- Attaching event listeners correctly to the hamburger menu, theme button, FAQ buttons, slider controls, notifications, and forms.
- Maintaining one common JavaScript file while safely handling pages that do not contain every interactive component.
- Implementing light/dark mode and storing the user's preference with localStorage.
- Managing relative paths between the html, css, js, and images folders so styling, scripts, and the logo load correctly.
- Making interactive components usable on smaller screens and adding accessibility attributes.
- Testing navigation, CSS loading, JavaScript behaviour, and each Practical 4 feature in the final working portal

F. Learning Outcomes:

Through this practical, I learned how JavaScript can be used with the DOM to create an interactive and user-friendly StudentHub portal. I gained practical experience in selecting DOM elements, changing classes and attributes, handling user events, and controlling dynamic content.

I also learned how to implement a collapsible FAQ, modal popup, content slider, notification banner, hamburger navigation, and light/dark theme switcher. Using localStorage helped me understand how UI preferences can be saved and restored when the user returns to the page.

The practical also improved my understanding of accessibility and responsive interaction.

I used ARIA attributes, keyboard-friendly modal closing, focus restoration, and responsive navigation so that dynamic features remain usable across the StudentHub pages.

Overall, this practical improved my understanding of JavaScript DOM manipulation, event handling, class manipulation, localStorage, basic animation, and client-side UI development.

Faculty Remarks: